

$^{34}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15**

- 1985Ra15:** 6×10^{11} n/cm²/s thermal neutrons at the target position were produced from the Los Alamos Omega West Reactor at a nominal 8-MW reactor power. γ rays following thermal neutron capture by a 1.09-g, 77.7% enriched ^{34}S target were detected using a 26 cm³ coaxial Ge(Li) detector inside a 20-cm-diam by 30-cm-long NaI(Tl) annulus with FWHM=2.3 keV at 1 MeV, 5.5 keV at 6 MeV and 8.8 keV at 11 MeV. Measured σ_γ , E_γ , and I_γ for 61 γ -ray transitions. Deduced 22 levels and γ -ray branching ratios.
- 1972Dz13:** Thermal neutrons were produced from the thermal column of the IRT reactor of the Baghdad Nuclear Research Institute. γ rays following thermal neutron capture by a 1.5-g, 97%-enriched ^{34}S target were detected using a 10-cm³ Ge(Li) detector with FWHM=3.5 keV at 1 MeV and 8 keV at 7 MeV. Measured E_γ and I_γ for 22 γ -ray transitions. Deduced 11 levels.
- 1985Ke08:** 5×10^{12} n/cm²/s thermal neutrons were produced from the McMaster University Reactor. γ rays following thermal neutron capture by a natural sulfur target were detected using a 10% efficient Ge centered within a NaI(Tl) annulus with FWHM=3.2-4.4 keV at 4-7 MeV. Measured E_γ and I_γ for 13 γ -ray transitions. Deduced 9 levels.
- 1997Be42:** Thermal neutrons were provided from the reactor BR1 of the Studiecentrum voor kernenergie in Mol, Belgium. γ rays following thermal neutron capture by a 313.6-mg, 93.26%-enriched ^{34}S and 5.933%-enriched ^{36}S target were detected using a Ge detector. Measured σ_γ , E_γ , and I_γ for 11 γ -ray transitions. Deduced 8 levels. Compared with direct-capture (DC) model calculations.
- All 55 E_γ values from **2007ChZX**: database of prompt gamma activation analysis (PGAA) from neutron capture are in agreement with the data below.

 ^{35}S Levels

E(level) [†]	Comments
0	
1572.374 7	
1991.28 4	
2347.781 8	C ² S=0.48 for 3/2 ⁻ from 1997Be42 .
2717.013 20	
2938.63 5	
3558.082 27	
3801.955 15	C ² S=0.08 for 3/2 ⁻ from 1997Be42 .
4105.6 8	
4189.242 33	C ² S=0.15 for 1/2 ⁻ from 1997Be42 .
4477.61 7	
4903.362 15	C ² S=0.49 for 1/2 ⁻ from 1997Be42 .
4963.083 25	C ² S=0.19 for 3/2 ⁻ from 1997Be42 .
5752.5 8	
6018.8 6	
6078.477 32	
6293.93 6	
6354.89 23	
6419.9 11	
6629.43 9	
6761.0 12	
6986.099 24	

[†] From a least-squares fit to γ -ray energies.

 $\gamma(^{35}\text{S})$

Photons per 100 thermal neutron captures are reported in **1972Dz13** and **1985Ke08**. γ -ray cross sections in mb are reported in **1985Ra15**, multiplying these by 0.340 yields the number of photons per 100 thermal neutron captures. $\sigma(4638)=163$ mb 15.
 $I_\gamma(4638)=\sigma(4638)*0.340=55$ 5.
 γ -ray intensities per incident neutron for all transitions and γ -ray cross sections in mb for primary transitions are reported in **1997Be42**. The strongest 4638 γ is used for normalization. $\sigma(4638)=148.6$ mb 55 and
 $I_\gamma(4638)=\sigma(\text{1997Be42})/\sigma(\text{1985Ra15})*I_\gamma(\text{1985Ra15})=50$ 7.

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$^{34}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{35}\text{S})$ (continued)

E_γ †	I_γ ‡	$E_i(\text{level})$	E_f	Comments
356.66 @ 9	0.037 @ 4	2347.781	1991.28	
356.66 @ 9	0.037 @ 4	6986.099	6629.43	
368.5 4	0.020 7	2717.013	2347.781	
619.23 19	0.042 8	3558.082	2938.63	
631.32 @ 24	0.060 @ 9	4189.242	3558.082	
631.32 @ 24	0.060 @ 9	6986.099	6354.89	
x 646.9 ‡ 5	0.7 1			E_γ, I_γ : from 1972Dz13. This γ ray has not been observed in any other measurements. 1972Dz13 assigned 6986 as the initial level.
x 663.41 ‡ 7	0.09 1			
692.16 5	0.133 17	6986.099	6293.93	
775.398 6	17.2 19	2347.781	1572.374	E_γ : other: 775.50 15 (1972Dz13). I_γ : weighted average of 19.5 20 (1972Dz13) and 15.6 17 (1985Ra15).
x 803.81 ‡ 9	0.092 14			
863.28 28	0.034 10	3801.955	2938.63	
907.608 ‡ 23	0.56 10	6986.099	6078.477	E_γ : weighted average of 908.1 3 (1972Dz13) and 907.607 14 (1985Ra15). I_γ : weighted average of 0.5 1 (1972Dz13) and 0.61 10 (1985Ra15).
1084.79 15	0.054 10	3801.955	2717.013	
1101.92 31	0.033 8	4903.362	3801.955	
1144.591 20	0.55 6	2717.013	1572.374	
1161.05 20	0.055 8	4963.083	3801.955	
1210.28 4	0.252 27	3558.082	2347.781	
1250.61 5	0.214 24	4189.242	2938.63	
x 1381.67 ‡ 24	0.024 8			
1404.967 24	0.60 6	4963.083	3558.082	
1454.09 4	0.45 11	3801.955	2347.781	
1566.7 3	0.47 9	3558.082	1991.28	
1572.333 8	35.0 34	1572.374	0	E_γ : weighted average of 1572.15 16 (1972Dz13) and 1572.333 7 (1985Ra15). I_γ : others: 39 4 (1972Dz13) and 32 4 (1997Be42).
1760.55 11	0.146 21	4477.61	2717.013	
1840.2 12	2.2 4	4189.242	2347.781	E_γ : unweighted average of 1839 1 (1972Dz13) and 1841.426 15 (1985Ra15). I_γ : weighted average of 3.6 8 (1972Dz13) and 2.14 21 (1985Ra15).
1964.8 2	0.37 10	4903.362	2938.63	
1991.27 5	0.54 6	1991.28	0	
2022.954 9	11 1	6986.099	4963.083	E_γ : others: 2022.80 20 (1972Dz13) and 2023.0 10 (1985Ke08). I_γ : weighted average of 12.0 15 (1972Dz13), 11 1 (1985Ke08), 11.4 10 (1985Ra15), and 9.1 12 (1997Be42).
2082.83 17	15.9 15	6986.099	4903.362	E_γ : unweighted average of 2082.65 16 (1972Dz13), 2083.17 7 (1985Ke08), and 2082.681 12 (1985Ra15). I_γ : weighted average of 18.0 15 (1972Dz13), 17 2 (1985Ke08), 15.6 17 (1985Ra15), and 12.6 17 (1997Be42).
2184.16 ‡ 19	4.7 5	4903.362	2717.013	E_γ, I_γ : from 1985Ke08. This γ ray has not been observed in any other measurements.
2229.510 16	2.86 27	3801.955	1572.374	
2347.700 15	49 4	2347.781	0	E_γ : weighted average of 2347.50 10 (1972Dz13), 2347.71 3 (1985Ke08), and 2347.701 11 (1985Ra15). I_γ : weighted average of 54 5 (1972Dz13), 49 4 (1985Ke08), 50 5 (1985Ra15), and 40 6 (1997Be42).
2508.39 8	0.38 4	6986.099	4477.61	
2555.492 14	3.14 31	4903.362	2347.781	E_γ : other: 2555.9 4 (1972Dz13). I_γ : weighted average of 3.4 7 (1972Dz13) and 3.09 31 (1985Ra15).
2615.2 2	1.09 10	4963.083	2347.781	E_γ : other: 2614.3 3 doublet (1972Dz13). I_γ : other: <1.3 4 (1972Dz13).

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$^{34}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{35}\text{S})$ (continued)

E_γ †	I_γ †#	$E_i(\text{level})$	E_f	Comments
2616.8 13	0.24 7	4189.242	1572.374	E_γ : other: 2614.3 3 doublet (1972Dz13). I_γ : other: <1.3 4 (1972Dz13).
2716.99 16	0.30 5	2717.013	0	
2796.74 4	4.9 4	6986.099	4189.242	E_γ : weighted average of 2796.83 13 (1985Ke08) and 2796.73 4 (1985Ra15). Other: 2796.6 5 (1972Dz13). I_γ : weighted average of 6.1 8 (1972Dz13), 4.3 4 (1985Ke08), 5.4 5 (1985Ra15), and 4.9 7 (1997Be42).
2905.1 4	0.14 4	4477.61	1572.374	
2938.58 11	0.89 9	2938.63	0	
2972.0 4	0.18 6	4963.083	1991.28	
3139.9 5	0.078 24	6078.477	2938.63	
3183.95 4	6.1 6	6986.099	3801.955	E_γ : weighted average of 3184.30 20 (1972Dz13), 3183.94 7 (1985Ke08), and 3183.94 4 (1985Ra15). I_γ : weighted average of 7.0 9 (1972Dz13), 5.7 6 (1985Ke08), 6.2 6 (1985Ra15), and 5.7 8 (1997Be42).
3330.84 13	7.6 7	4903.362	1572.374	E_γ : unweighted average of 3331.08 15 (1972Dz13), 3330.64 6 (1985Ke08), and 3330.80 4 (1985Ra15). I_γ : weighted average of 7.8 10 (1972Dz13), 7.7 8 (1985Ke08), and 7.4 7 (1985Ra15).
3390.62 7	5.6 5	4963.083	1572.374	E_γ : weighted average of 3390.86 20 (1972Dz13), 3390.73 7 (1985Ke08), and 3390.55 5 (1985Ra15). I_γ : weighted average of 5.5 8 (1972Dz13), 5.7 6 (1985Ke08), and 5.5 5 (1985Ra15).
3558.1 5	0.085 17	3558.082	0	
3801.73 4	2.59 23	3801.955	0	E_γ : weighted average of 3802.0 3 (1972Dz13), 3801.49 14 (1985Ke08), and 3801.74 3 (1985Ra15). I_γ : weighted average of 3.6 7 (1972Dz13), 2.3 2 (1985Ke08), 3.09 31 (1985Ra15), and 2.6 4 (1997Be42).
4105.3 8	0.048 17	4105.6	0	
4188.95 5	2.54 27	4189.242	0	E_γ : weighted average of 4188.80 13 (1985Ke08) and 4188.96 4 (1985Ra15). Other: 4189.6 3 (1972Dz13). I_γ : weighted average of 1.9 4 (1972Dz13), 2.72 27 (1985Ra15), and 2.8 4 (1997Be42). Other: 24 2 (1985Ke08).
4268.65 14	0.34 4	6986.099	2717.013	E_γ : other: 4268.4 7 (1972Dz13). I_γ : other: 0.4 2 (1972Dz13).
4638.14 14	56 5	6986.099	2347.781	E_γ : unweighted average of 4638.4 2 (1972Dz13), 4638.10 3 (1985Ke08), and 4637.91 4 (1985Ra15). I_γ : weighted average of 56 6 (1972Dz13), 59 5 (1985Ke08), 55 5 (1985Ra15), and 50 7 (1997Be42).
4902.99 5	4.0 4	4903.362	0	E_γ : weighted average of 4903.07 7 (1985Ke08) and 4902.96 4 (1985Ra15). Other: 4903.4 3 (1972Dz13). I_γ : weighted average of 3.2 8 (1972Dz13), 4.3 4 (1985Ke08), 3.7 4 (1985Ra15), and 4.2 6 (1997Be42).
4963.06 22	2.62 27	4963.083	0	E_γ : unweighted average of 4963.2 5 (1972Dz13), 4963.34 10 (1985Ke08), and 4962.63 5 (1985Ra15). I_γ : weighted average of 2.6 8 (1972Dz13), 2.92 27 (1985Ra15), and 2.18 33 (1997Be42). Other: 29 3 (1985Ke08).
5752.0 8	0.018 5	5752.5	0	
6018.2 6	0.020 7	6018.8	0	
6077.87 11	0.35 4	6078.477	0	E_γ : other: 6078.2 10 (1972Dz13). I_γ : weighted average of 0.3 1 (1972Dz13) and 0.36 4 (1985Ra15).
6293.2 4	0.065 14	6293.93	0	
6355.0 6	0.046 7	6354.89	0	
6419.3 11	0.016 5	6419.9	0	
6628.5 6	0.030 6	6629.43	0	

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³⁴S(n,γ) E=thermal **1985Ra15** (continued)

γ(³⁵S) (continued)

<u>E_γ[†]</u>	<u>I_γ^{†#}</u>	<u>E_i(level)</u>	<u>E_f</u>
6760.3 12	0.019 8	6761.0	0
6985.7 10	0.036 8	6986.099	0

[†] From **1985Ra15**, unless otherwise noted.

[‡] Not observed in **2007ChZX** (PGAA).

Intensity per 100 neutron captures.

@ Multiply placed with undivided intensity.

^x γ ray not placed in level scheme.

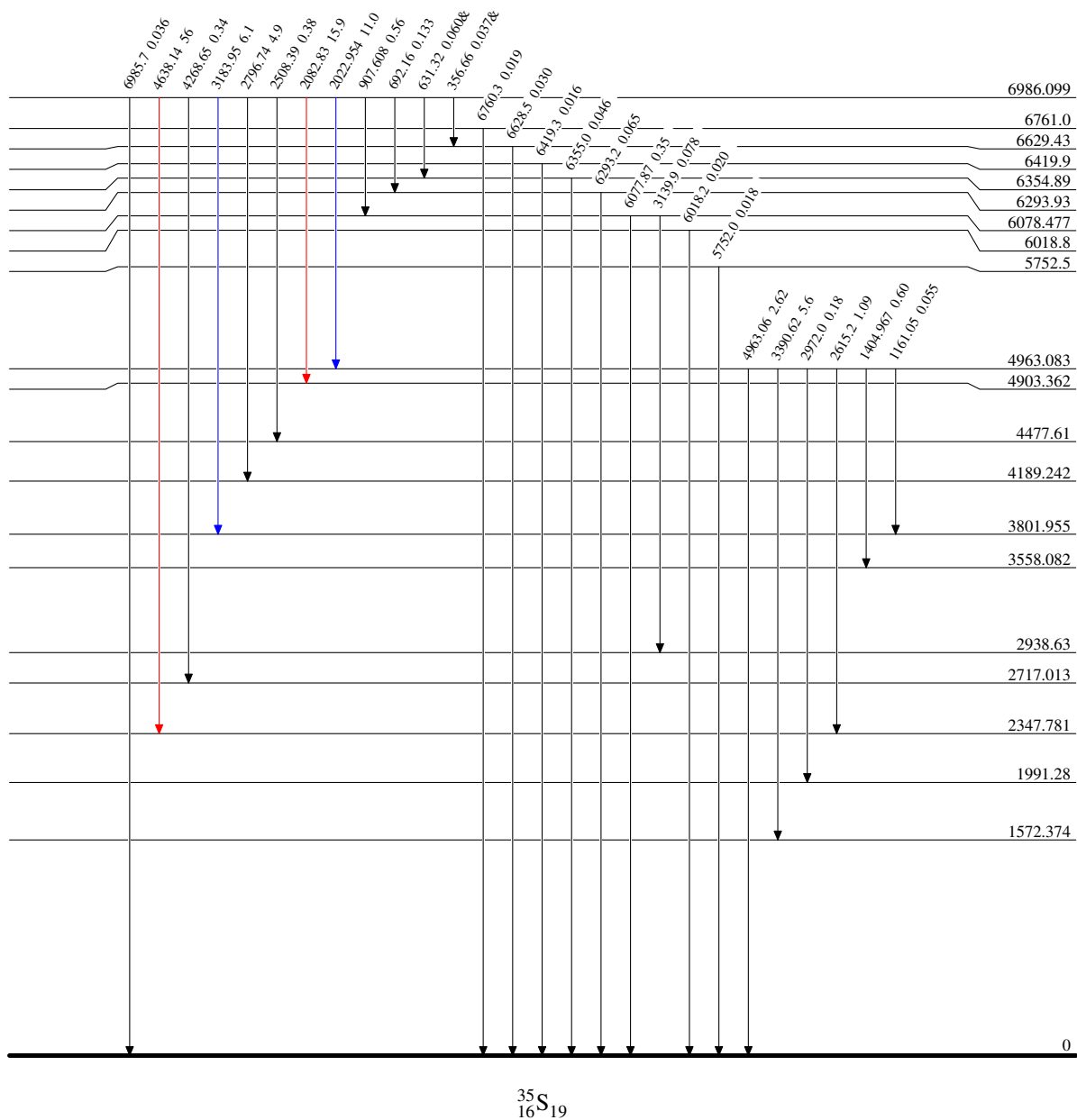
$^{34}\text{S}(\text{n},\gamma) \text{E=thermal}$ 1985Ra15

Level Scheme

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

Legend

$\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$



³⁴S(n,γ) E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

Legend

$$\longrightarrow \mathbf{I}_\gamma < 2\sigma \times \mathbf{I}_\gamma^{max}$$
$$\mathbf{I}_\gamma < 10\% \times \mathbf{I}_\gamma^{max}$$

$\mathbf{I}_\gamma > 10\% \times \mathbf{I}_\gamma^{max}$

